Integrated UAV and Satellite Observations for Antarctic Melt Pond Monitoring: Assessing the Melt-Induced Risk for Transportation on Landfast Ice

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***Abstract*—In recent years, the extent of Antarctic sea ice has frequently reached historically low levels. Melt ponds significantly affect the sea ice heat balance and mass balance while posing challenges to ice surface transportation. Melt ponds in the Arctic have been extensively monitored using satellite remote sensing. However, the Antarctic has received very little attention, and small-scale melt pond distribution patterns remain virtually unexplored. This study combines high-resolution uncrewed aerial vehicle (UAV) imagery with satellite remote sensing to investigate the characteristics of melt ponds on landfast ice near Zhongshan Station, East Antarctica. UAV-derived data reveal that the melt ponds in this region are small (median area = 0.34 m^2) and irregular in shape (mean roundness = 4.97). Validation against the UAV data suggests poor performance of the traditional melt pond fraction (MPF) retrieval model in the Antarctic region. Therefore, we develop an MPF retrieval algorithm by integrating UAV and Landsat 8 images. The correlation coefficient is as high as 0.82, substantially outperforming the traditional method. Dynamic MPF monitoring indicates that melt ponds concentrate near the coastline, especially to the west of bare rock areas, as the persistent summer easterly winds transport low-albedo dust westward. The integrated melt-induced risk assessment, combining MPF and Normalized Difference Water Index for ice, shows that high-risk areas are concentrated in Nella Fjord, west of the Mirror Peninsula. Ice surface transport should avoid this area and be completed before mid-November.**

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***Index Terms*—Antarctic, melt ponds, remote sensing, risk assessment.**

I. Introduction

As A primary cold source within the climate system, Antarctica significantly influences global climate change and ranks among the crucial regions for predicting future shifts in global climate [1]. Sea ice, a critical component of polar environments, significantly influences the global exchange of heat and mass. Due to its high albedo, sea ice significantly limits the absorption of incoming solar shortwave radiation by the ocean surface. This suppression of energy input weakens heat and moisture exchange between the ocean and atmosphere, thereby modifying the surface energy balance of the ocean [2], [3]. Furthermore, the melting or freezing of sea ice alters local seawater salinity, disrupts ocean stratification, and impacts the Antarctic Intermediate Water and Antarctic Bottom Water [4]. Over a longer period, the extent of Antarctic sea ice showed a growth trend that was different from that of the Arctic. However, this trend has reversed in recent years (2016–2024), during which unusually low sea ice extent has been observed [5]. Currently, the mechanism underlying the changes in Antarctic sea ice following 2016 and its implications for future trends remain elusive [6].

Meltwater from sea ice usually accumulates in melt ponds in the summer and can cover up to 50%–60% of the ice surface [7], [8]. Because the albedo of melt ponds is lower than that of the surrounding ice, more solar radiation is absorbed, which further enhances ice melt [9], [10], [11]. Furthermore, as melt ponds evolve, they alter the permeability of sea ice, thereby influencing the heat budget within the ocean mixed layer [12]. Enhanced brine drainage beneath melt ponds facilitates salt removal from sea ice, while the latent heat released during freeze-up impacts the basal growth of sea ice. Consequently, the dynamics of melt ponds play a vital role in regulating the heat balance and mass balance of sea ice, thereby influencing polar climate change.

Each summer, Antarctic research stations rely on large icebreakers to transport supplies. For example, research stations located in Prydz Bay, such as Zhongshan Station, Bharati Station, and Progress Station, depend on this logistical support.

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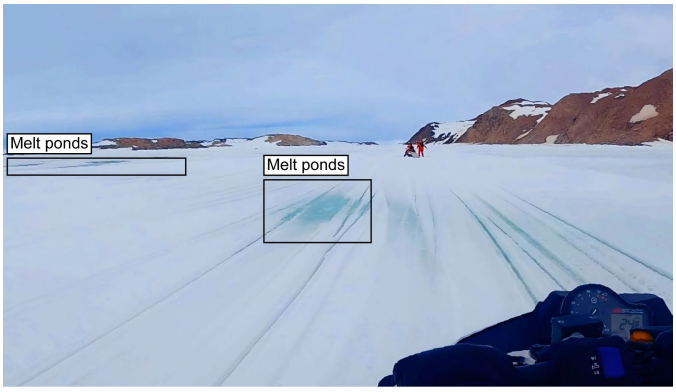


Fig. 1. Melt ponds appear on the landfast ice along transportation and field activity routes near Zhongshan Station. Image credit: Lei Zheng, December 17, 2021. The black-bordered rectangles mark the melt ponds.

However, due to the ships' draft limitations, icebreakers cannot directly access these stations, making it necessary to rely on ice surface transport for the delivery of supplies and personnel. During this period, extensive surface melting of landfast ice results in the formation of melt ponds (see Fig. 1), which may pose challenges to ice surface transportation and field activities [13], [14]. Therefore, monitoring melt ponds on the ice surface is crucial for ensuring the safety of ice-based operations.

Satellite remote sensing enables large-scale monitoring of melt ponds on sea ice. Although microwave remote sensing, particularly synthetic aperture radar (SAR), is largely unaffected by weather and has made some progress in melt pond monitoring, its application in high-precision melt pond fraction (MPF) retrieval remains limited. Factors such as surface conditions, roughness, and wind limit its accuracy and generalizability, and it is not yet a primary monitoring approach [15], [16], [17], [18], [19], [20], [21], [22]. In contrast, optical satellite remote sensing (such as MODIS, Landsat, and Sentinel-2) provides more direct observations of melt ponds through their distinct spectral signatures, especially their low reflectance in the visible and near-infrared bands relative to surrounding ice and snow. Numerous efforts have been made to estimate MPF from optical satellite imagery and to uncover the physical mechanisms underlying its variation. These efforts can be broadly categorized into three groups based on their fundamental principles [23], [24], [25], [26], [27]. One approach involves retrieving MPF using traditional statistical methods, including the linear spectral unmixing (LSU) algorithm [23], [24], [25] and the linear polar (LP) algorithm [28], [29]. An alternative approach widely adopted derives MPF and surface albedo through physical modeling [30], [31]. The application of artificial intelligence, particularly machine learning and deep learning, has expanded rapidly in remote sensing monitoring over the past few years. Several studies have explored this approach for melt pond identification and MPF retrieval [24], [26], [32], [33], [34].

However, satellite imagery with spatial resolutions in the range of tens to hundreds of meters is insufficient to accurately capture the shape and distribution of melt ponds. With high flexibility, uncrewed aerial vehicle (UAV) remote sensing provides high-resolution data, effectively complementing mesoscale observations that lie between satellite and ground-based measurements [35]. UAVs are capable of capturing detailed features of melt ponds and have proven effective for small-scale melt pond investigations and algorithm validation [36], [37], [38].

Unlike the rapidly diminishing Arctic sea ice, the melting of Antarctic sea ice has received comparatively limited attention. The above endeavors related to remote sensing-based melt pond monitoring are predominantly focused on Arctic sea ice. These algorithms have yet to be applied in the Antarctic region, and their accuracy and efficacy remain unconfirmed. Moreover, there is a lack of research on the morphology and distribution of melt ponds at a fine scale. This poses a significant challenge to our understanding of Antarctic sea ice changes and to ensuring the safety of ice transportation. In this study, we initially analyze the distribution pattern and morphological characteristics of melt ponds using high spatial resolution imagery captured by UAVs. We then evaluate and develop remote sensing algorithms for MPF retrieval by integrating data from UAV imagery and satellite imagery. Lastly, the MPF of the landfast ice at Zhongshan Station is dynamically monitored, and the potential impacts of surface melt intensity on ice operations are assessed.

II. Study Region and Dataset

A. Study Region

This study primarily focuses on the development of melt ponds on landfast ice adjacent to Zhongshan Station [see Fig. 2(a)]. Zhongshan Station, the largest Antarctic research base currently operated by China, is located on the Mirror Peninsula in the Larsemann Hills of East Antarctica (69°22′S, 76°22′E), where the coastal area remains covered by landfast ice for most of the year. The Chinese Antarctic Research Expedition (CHINARE) and RV Xuelong usually visit and resupply Zhongshan Station from November to December each year [39]. During this period, the offshore extent of landfast ice ranges from 20 to 40 km, with the maximum ice thickness being approximately 1.8 m [40]. However, due to the ships' draft and icebreaking limitations, the icebreaker cannot directly access Zhongshan Station. Containers loaded with various supplies, including food, fuel, and equipment, are first offloaded onto the nearshore landfast ice, and the supplies are then transported to Zhongshan Station using vehicles such as snowmobiles and helicopters [14], [41], [42], [43]. Meanwhile, rising summer temperatures lead to significant melting of the landfast ice, resulting in melt ponds of varying sizes scattered across the region, posing risks to transportation. Therefore, continuous monitoring of melt ponds in the region is crucial, as it is one of the key measures to ensure the safety of operations on the ice.

B. UAV Data

Three UAV surveys of landfast ice melt ponds near Zhongshan Station were conducted during the 34th and 38th CHINARE. The UAV data acquisition was carried out using the FEIMA F1000 medium-sized fixed-wing

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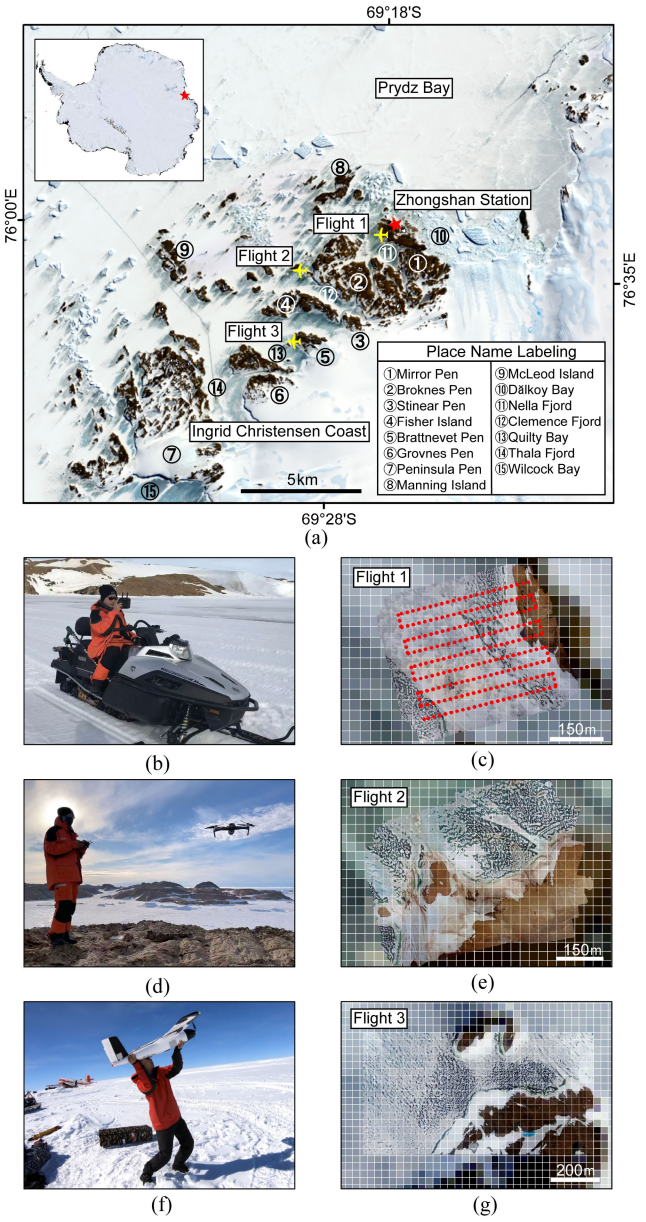


Fig. 2. Location and images of the study region. (a) Extent of the study region. The star indicates the location of Zhongshan Station, and the yellow airplane symbol marks the site where UAV data were collected. (b) UAV data collection was conducted on December 16, 2021. (c) UAV observations on December 16, 2021. The background image is a Landsat 8 true-color composite acquired on December 17, 2021. Red dots indicate an example of the UAV flight track. The white grids represent the Landsat pixel layout. (d) UAV data collection was conducted on December 20, 2021. (e) UAV observations on December 20, 2021. The background image is a Landsat 8 true-color composite acquired on December 19, 2021. (f) UAV data collection was conducted on January 17, 2018. (g) UAV observations on January 17, 2018. The background image is a Landsat 8 true-color composite acquired on January 16, 2018.

UAV (during the 34th CHINARE) and the DJI Mavic 2 Pro small multirotor UAV (during the 38th CHINARE). The two UAVs were equipped with a Sony α5100 digital camera with a sensor size of 23.5 $\times$ 15.6 mm and a Hasselblad L1D-20c gimbal camera with a sensor size of 12.825 $\times$ 8.550 mm, respectively. Both UAVs had built-in GPS receivers, which provided spatial coordinates during imaging. On January 17, 2018, and December 16 and 20, 2021, three UAV flights were conducted. The corresponding flight locations are illustrated in Fig. 2. Detailed flight parameters of the UAVs are presented in Table I.

C. Satellite Data

Landsat 8 Operational Land Imager (OLI) data provide 30 m resolution multispectral imagery with a 16-day revisit cycle, making them appropriate for monitoring MPF on a large scale [44]. The Landsat 8 Level-2 surface reflectance product, atmospherically corrected using the Land Surface Reflectance Code (LaSRC, Collection 2), is used in this study. The LaSRC algorithm performs atmospheric correction based on the Second Simulation of the Satellite Signal in the Solar Spectrum–Vector (6SV) radiative transfer model, accounting for Rayleigh scattering, gaseous absorption, and aerosol effects. All Landsat 8 images used in this study during the austral summer (from November 1 to March 1) were manually inspected for usability (see Table II). Notably, although Sentinel-2 provides higher spatial resolution than Landsat 8, no usable satellite images were available during the periods when UAV images were acquired. This prevented algorithm evaluation and development based on Sentinel-2 imagery and limited improvements in monitoring frequency.

D. Auxiliary Data

We used data from the automatic weather station at Zhongshan Station in order to investigate how meteorological variables affect the spatiotemporal evolution of melt ponds. The station recorded data including air temperature, air pressure, relative humidity, wind conditions, and cloud cover, with observations taken every six hours [45]. The air temperature, wind speed, and wind direction records used for this study were collected from November 1, 2021, through March 1, 2022. The surface spectral information was obtained during the 36th CHINARE using a spectroradiometer manufactured by Analytical Spectral Devices (ASD) to support the investigation of the influence of surface spectral characteristics on the spatial distribution of melt ponds. Measurements were conducted near Zhongshan Station (69°22′12.49″S, 76°21′49.28″E). Under favorable weather conditions with clear skies and few clouds on December 28, 2019, spectral reflectance data (350–2500 nm) of bare rock, clean snow, and snow affected by dust were collected. To delineate the monitoring area for melt ponds on the ice surface, a high-resolution coastline shapefile provided by the Antarctic Digital Database (https://add.scar.org) and the Landsat Image Mosaic of Antarctica (LIMA), available through the United States Geological Survey (https://lima.usgs.gov), were used.

III. Methodology

We employ the following methods to monitor melt ponds in the Zhongshan Station area of Antarctica at multiple spatial scales.

1) Orthophotos are generated from UAV imagery to extract high-precision melt pond parameters, analyze the distribution patterns of small-scale melt ponds, and provide

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TABLE I
FLIGHT PARAMETERS OF THE UAVS

Parameter name	Flight 1,2	Flight 3
Flying height	100 m	370 m
Ground Sample Distance	0.05 m	0.11 m
Image coverage	27.36 × 18.24 m	66 × 44 m
Oblique angle	90°	90°

TABLE II
LANDSAT 8 IMAGERY USED IN THIS STUDY

Scene ID	Date	Path	Row	Sun elevation	Sun azimuth	Cloud cover
LC08_L2SR_125109_20180116_20201016_02_T2	January 16, 2018	125	109	32.39	60.55	3.47
LC08_L2SR_124109_20211108_20211117_02_T2	November 8, 2021	125	109	30.16	52.11	0.00
LC08_L2SR_125109_20211117_20211125_02_T2	November 17, 2021	124	109	32.37	53.25	0.00
LC08_L2SR_125109_20211124_20211201_02_T2	November 24, 2021	125	109	33.74	54.26	0.00
LC08_L2SR_126109_20211201_20211209_02_T2	December 1, 2021	126	109	34.76	55.35	0.00
LC08_L2SR_126109_20211217_20211223_02_T2	December 17, 2021	126	109	35.71	57.84	52.76
LC08_L2SR_124109_20211219_20211223_02_T2	December 19, 2021	124	109	35.69	58.12	0.00
LC08_L2SR_126109_20220102_20220106_02_T2	January 2, 2022	126	109	34.71	59.75	0.00
LC08_L2SR_124109_20220104_20220113_02_T2	January 4, 2022	124	109	34.46	59.92	0.00
LC08_L2SR_125109_20220212_20220222_02_T2	February 12, 2022	125	109	25.20	59.10	15.47
LC08_L2SR_126109_20220219_20220301_02_T2	February 19, 2022	126	109	22.93	58.20	0.34

reference data for MPF algorithm development and validation.

2) MPF is retrieved from Landsat 8 images using the traditional LSU algorithm, with accuracy assessed against high-resolution reference values derived from UAV data.
3) A random forest regression (RFR) model is developed and evaluated by integrating UAV and Landsat 8 data, enabling melt pond monitoring across the Antarctic region at a large scale.

A. UAV Orthophoto Generation and Melt Pond Mapping

Melt pond orthophotos are created by applying structure-from-motion (SfM) and multi-view stereo (MVS) photogrammetric techniques [46]. The generation process, as shown in Fig. 3, is carried out using Agisoft Metashape Professional version 1.8.5. Orthophotos of the corresponding areas are generated from high-overlap UAV images, resulting in two products with a spatial resolution of 5 cm for Flights 1 and 2 and one product with a spatial resolution of 11 cm for Flight 3.

A pixel-based supervised classification procedure is performed on the orthophotos using a maximum likelihood classifier to extract information about melt ponds. The surface observed in the orthophotos is classified into three types: ice, melt ponds, and bare land. The red, green, and blue spectral bands serve as input features, with the classifier trained on representative samples from each category. When the data are normally distributed, the maximum likelihood is formulated as follows [31]

$$D = \ln(p_c) - [0.5 \ln(|\mathrm{Cov}_c|)] - \left[0.5 (X - H_c)^{\mathrm{T}} \left(\mathrm{Cov}_c^{-1}\right) (X - H_c)\right] \quad (1)$$

where D is the quantity-weighted distance (likelihood), c is a particular class, p_c is the priori probability of class c, Cov_c corresponds to the within-class covariance matrix for class c, while X and H_c denote the pixel's measurement vector and the class mean vector, respectively. After classifying images into three separate categories using the maximum likelihood classifier, the ice and bare land categories are combined into a single class labeled as "other" (see Fig. 3). An overall accuracy

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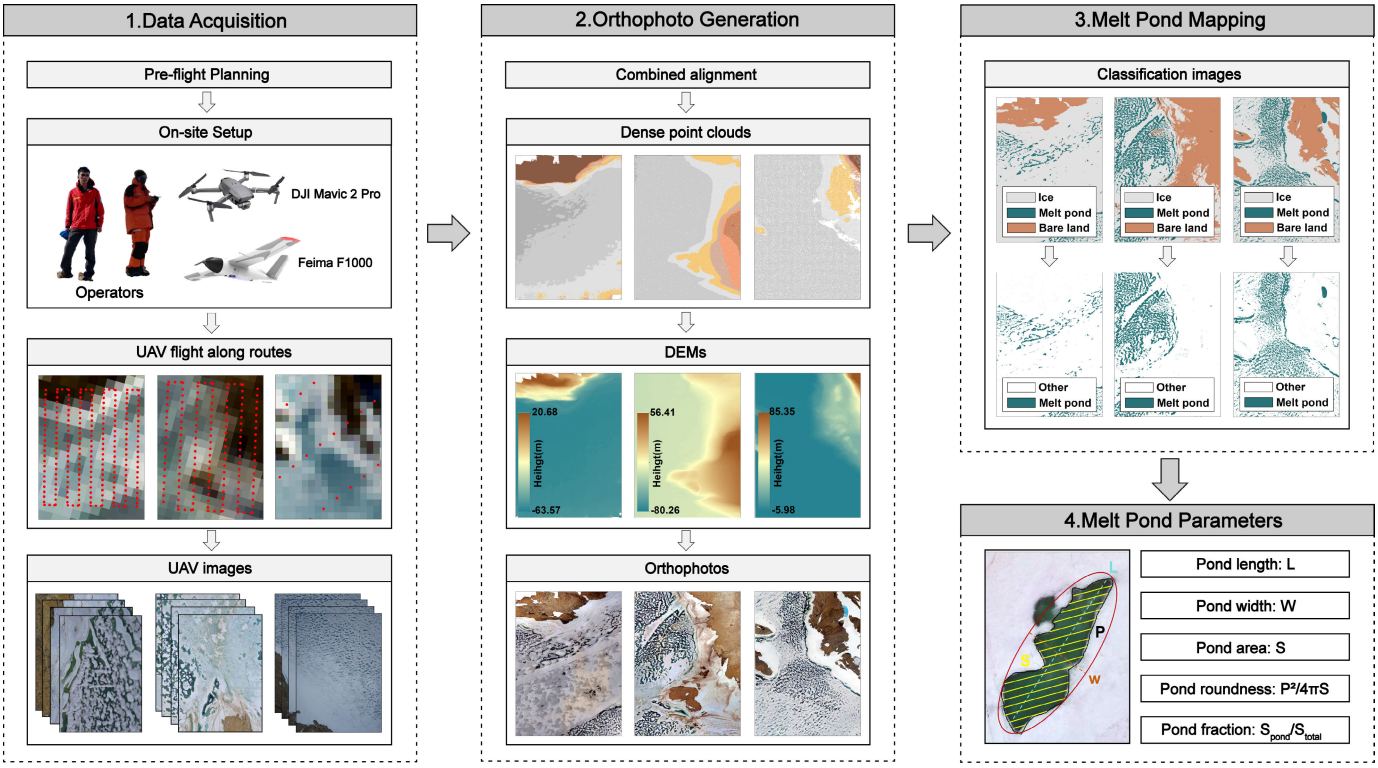


Fig. 3. Workflow of UAV data acquisition and processing.

exceeding 90% was obtained according to the validation results, indicating good classification performance.

The MPF derived from UAV data can be obtained by spatially and temporally co-registering the melt pond classification images with the corresponding Landsat 8 imagery (see Fig. S1), followed by resampling to a spatial resolution of 30 m using mean aggregation. The morphological parameters of melt ponds, including length, width, area, and roundness, can also be statistically analyzed to understand the properties of melt ponds at a fine scale [47]. An ellipse fitting method is employed to derive the major and minor axes of irregularly shaped melt ponds. The principle is as follows:

$$a_1x^2 + a_2xy + a_3y^2 + a_4x + a_5y + a_6 = 0 \tag{2}$$

where a is the coefficient of the ellipse equation. For each point (x, y), the equation can be written as a linear system

$$\begin{bmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_n^2 & x_ny_n & y_n^2 & x_n & y_n & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{3}$$

To guarantee that the fit conic is an ellipse, a specific constraint is applied

$$a_2^2 - 4a_1a_3 > 0. \tag{4}$$

The ellipse parameters in (2) are solved via least squares fitting, enabling the computation of the major and minor axes. The melt pond area is obtained by summing the areas of the pixels in this category. The roundness R is defined as follows:

$$R = \frac{P^2}{4\pi S} \tag{5}$$

where P is the perimeter, and S is the area. For a perfect circle, the R value is 1. The roundness value deviates further from 1 as the shape becomes more elongated [48].

B. MPF Retrieved by the LSU Algorithm

Landsat imagery is used to obtain regional melt pond information. First, we evaluate the most commonly used LSU algorithm based on the MPF data obtained by the UAV. As a classic algorithm for retrieving MPF, it has been widely applied in the Arctic region [23], [24], [27], [32]. The fundamental principle of the LSU algorithm is that each pixel in a remote sensing image may contain one or more surface types, and its reflectance is regarded as a linear combination of the fractions of these surface types and the corresponding reflectance values of their pure endmembers [31]. Based on actual observations in the Antarctic region, surface features in the study area are classified into three categories (ice, melt ponds, and bare land), and the LSU model is expressed as follows [23], [49]:

$$\begin{cases} A_{\mathrm{I}}r_{\mathrm{I}}(\lambda_{\mathrm{i}}) + A_{\mathrm{M}}r_{\mathrm{M}}(\lambda_{\mathrm{i}}) + A_{\mathrm{L}}r_{\mathrm{L}}(\lambda_{\mathrm{i}}) + n_{\mathrm{i}} = R(\lambda_{\mathrm{i}}) \\ A_{\mathrm{I}}r_{\mathrm{I}}(\lambda_{\mathrm{j}}) + A_{\mathrm{M}}r_{\mathrm{M}}(\lambda_{\mathrm{j}}) + A_{\mathrm{L}}r_{\mathrm{L}}(\lambda_{\mathrm{j}}) + n_{\mathrm{j}} = R(\lambda_{\mathrm{j}}) \\ A_{\mathrm{I}}r_{\mathrm{I}}(\lambda_{\mathrm{k}}) + A_{\mathrm{M}}r_{\mathrm{M}}(\lambda_{\mathrm{k}}) + A_{\mathrm{L}}r_{\mathrm{L}}(\lambda_{\mathrm{k}}) + n_{\mathrm{k}} = R(\lambda_{\mathrm{k}}) \end{cases} \tag{6}$$

where i, j, and k represent the red, blue, and near-infrared bands, respectively, which are selected based on spectral differences among surface types and the results of performance tests [23], [24], [27]. I, M, and L represent ice, melt pond, and bare land, respectively. A represents the abundance of each surface feature. $r(\lambda)$ denotes the reflectance of pure endmembers for each surface feature in the corresponding spectral band. $R(\lambda)$ is the reflectance value of the pixel. n is noise. Endmember selection follows strict and consistent

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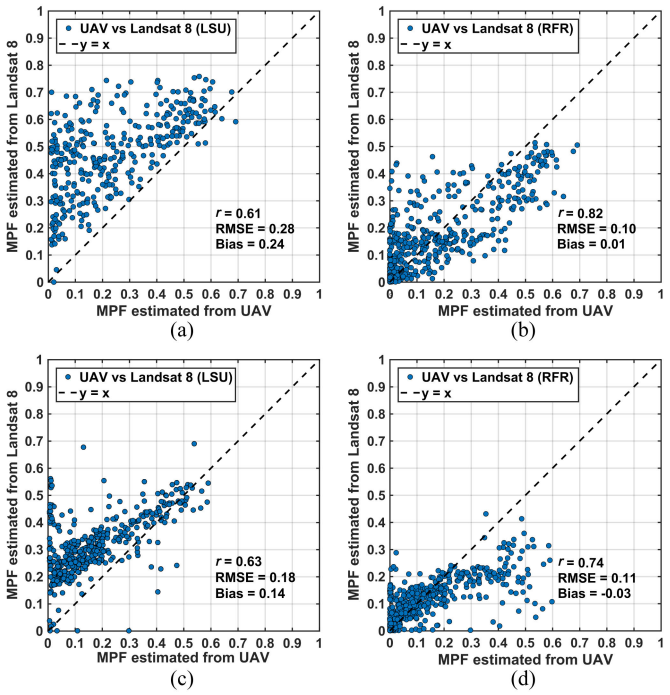


Fig. 4. Validation of MPF determined from Landsat 8 against that obtained from the UAV. The black line shows a 1:1 relation. (a) Comparison between MPF generated from the UAV (Flights 1 and 2) and MPF estimated from Landsat 8 images (December 17 and 19, 2021) by the LSU algorithm. (b) Fivefold cross-validation comparison between MPF generated from the UAV (Flights 1 and 2) and MPF estimated from Landsat 8 images (December 17 and 19, 2021) by the RFR algorithm. (c) Comparison between UAV-derived MPF from the independent validation flight (Flight 3) and MPF estimated from Landsat 8 images (January 17, 2018) by the LSU algorithm. (d) Comparison between UAV-derived MPF from the independent validation flight (Flight 3) and MPF estimated from Landsat 8 images (January 17, 2018) by the RFR algorithm.

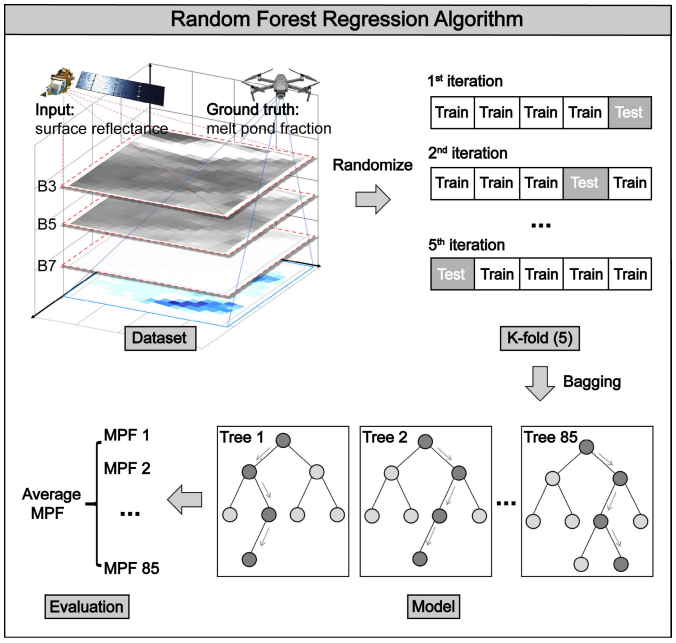


Fig. 5. RFR model used for MPF retrieval.

criteria to ensure that the extracted spectra reliably represent the true surface spectral characteristics. Endmember pixels are manually selected from spatially homogeneous and contiguous regions dominated by a single surface type, including sea ice, melt ponds, and bare land, while avoiding mixed pixels at the boundaries between different surface classes. To further reduce the influence of local variability and measurement noise, 30–50 representative pixels are selected for each endmember class, and the mean reflectance spectrum of these pixels is used as the final endmember reflectance spectrum.

The abundance sum-to-one constraint and abundance nonnegativity constraint are imposed to ensure that the inversion process is physically meaningful

$$\begin{cases} A_{I/M/L} > 0 \\ A_I + A_M + A_L = 1. \end{cases} \tag{7}$$

The above LSU model is overdetermined, as the number of equations exceeds the number of unknowns. Consequently, this problem is treated as an optimization task, in which the optimal solution is obtained through the least squares method [16]. With this approach, we obtain the MPFs of the study region from Landsat 8 images.

Based on Landsat 8 imagery, we evaluate the applicability of the LSU algorithm for retrieving MPF in Antarctica. Regarding algorithmic stability, variations in endmember reflectance between different periods (mid-December versus mid-January) have a minor influence on retrieval performance [see Fig. 4(a) and (c)]. We assess the sensitivity of the LSU algorithm to endmember uncertainty by perturbing the reflectance values of the ice, bare land, and melt pond endmembers. The results indicate that the correlation coefficient (r) varies by less than 0.1 (see Text S1 and Fig. S2). Although the LSU algorithm shows stability within the study region, it substantially overestimates MPF relative to UAV-derived values. In both tests, the r values are below 0.65, and the root mean square errors (RMSEs) reaches 0.28 and 0.18, indicating considerable discrepancies between the estimated and reference datasets [see Fig. 4(a) and (c)]. The LSU algorithm, therefore, fails to achieve sufficient retrieval accuracy, limiting its applicability in the study region.

C. MPF Retrieved by Machine Learning Model

Fig. 4(a) and (c) suggest that the retrieval accuracy of the classic LSU algorithm is unsatisfactory in MPF estimation over the landfast ice region of Prydz Bay. Therefore, we develop MPF retrieval models for this specific region using the MPF derived from high-resolution UAV data. In this study, we evaluate several machine learning regression algorithms for MPF estimation, including support vector regression, convolutional neural networks, Bayesian neural networks, and RFR. After repeated tests and evaluations, the RFR model is selected, and the surface reflectance from the green (B3), near-infrared (B5), and shortwave infrared 2 (B7) bands of Landsat 8 is used as input, achieving optimal accuracy while ensuring computational efficiency (see Fig. 5).

The RFR model improves prediction accuracy and model robustness by integrating multiple decision trees, making it a typical ensemble learning technique. By using bootstrap sampling, it generates multiple subsets of the dataset, each subset serving as the basis for training a separate decision tree. The final prediction is derived by aggregating the outputs of

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TABLE III
SUMMARY STATISTICS OF MELT POND PARAMETERS

Statistic	Length (m)	Width (m)	Area (m^2)	Roundness
Maximum	798.55	259.19	39519.27	2469.46
Mean	4.19	1.58	10.99	4.97
Median	1.38	0.73	0.34	3.18
Standard Deviation	16.46	4.06	382.51	29.27

all decision trees through averaging for regression tasks [50]

$$Y = \frac{1}{W}\sum_{w=1}^{W} T_w(\boldsymbol{B}) \tag{8}$$

where Y denotes the predicted output value. W is the total number of regression trees in the forest. $T_w(\boldsymbol{B})$ represents the prediction given by the wth regression tree for the input. $\boldsymbol{B}$ is the input feature vector. The RFR model is constructed with surface reflectance data as inputs and MPF as the output in this study. The number of regression trees is 85 (see Fig. 5). Data from Flights 1 and 2 are used for model development and testing, whereas data from Flight 3 are used for independent validation of the model. Since no suitable Landsat 8 images were available for the study region on the UAV observation dates (January 17, 2018, and December 16 and 20, 2021), images from January 16, 2018, and December 17 and 19, 2021 are selected. Fivefold cross-validation is used for a preliminary evaluation of the RFR model. The evaluation is based on the mean performance across all folds, offering a balanced assessment of the algorithm's potential [51]. In addition, data from Flight 3 are used to evaluate the model's practical retrieval performance, further examining its generalization ability.

Validation using UAV-derived MPF demonstrates that the RFR algorithm significantly outperforms the LSU method [see Fig. 4(a)], with r, RMSE, and bias from cross-validation reaching 0.82, 0.10, and 0.01, respectively [see Fig. 4(b)]. In addition, despite notable spatiotemporal heterogeneity between the independent validation data and the training dataset, and differences in the solar elevation angles of the Landsat 8 imagery used for retrieval (see Table II), r remains high at 0.74, and the RMSE is only 0.11 [see Fig. 4(d)], indicating that the RFR model demonstrates good generalization ability. The above validation results demonstrate that the proposed algorithm can efficiently and accurately retrieve MPF in the landfast ice region near Zhongshan Station, thereby meeting the monitoring requirements.

IV. RESULT

A. Morphology of Melt Ponds From UAV Observations

Using the maximum likelihood classifier, a total of 12 398 melt ponds are identified from three UAV orthophotos. The distributions of pond length and width are shown in Fig. 6(a) and (c), respectively. The mean length is roughly three times greater than the mean width, as summarized in Table III. A total of 3 907 ponds have lengths less than 1 m, making up roughly one-third of the total. In contrast, two-thirds of the melt ponds have widths of less than 1 m. This discrepancy between length and width suggests that the melt ponds in the region are generally irregular in shape rather than near-circular. This is also confirmed in Fig. 6(b) and (d). A total of 10 586 melt ponds have lengths of no more than 5 m, accounting for 85% of all ponds, while 11 974 ponds have widths of no more than 5 m, representing 90% of the total. Less than 1% of melt ponds have lengths exceeding 50 m, highlighting that small melt ponds dominate in this region. We use a power-law function to fit the probability density distributions of the length and width, reflecting a strong degree of correlation [see Fig. 6(a) and (c)].

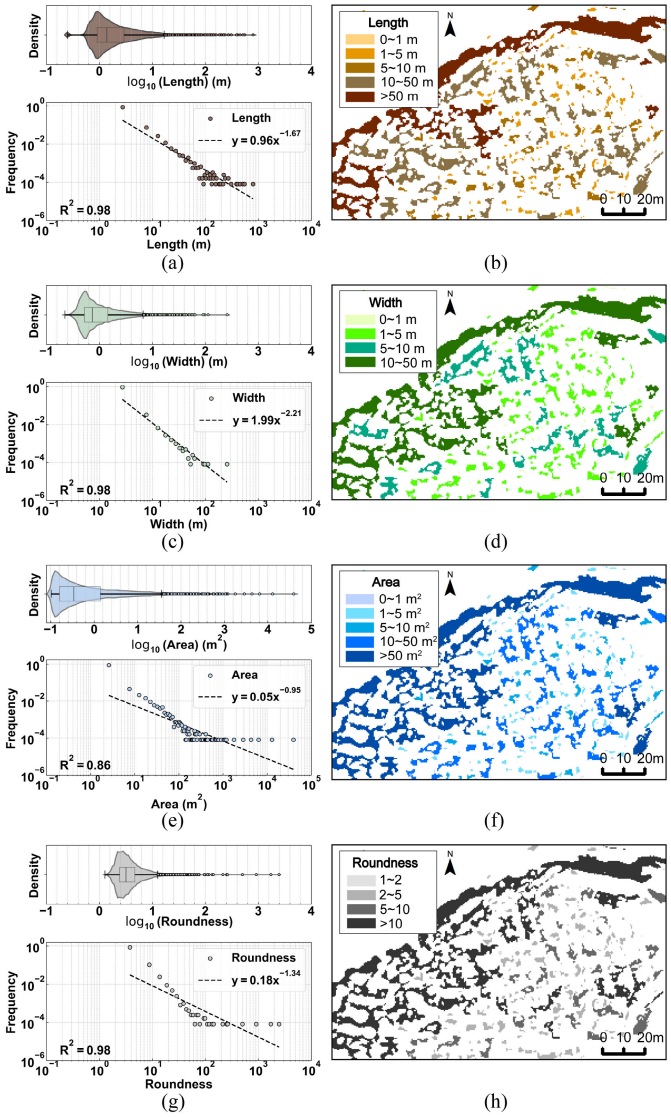


Fig. 6. Melt pond parameters obtained from UAV images. (a), (c), (e), and (g) Probability density distributions of length, width, area, and roundness. (b), (d), (f), and (h) Graded maps of length, width, area, and roundness in a partially enlarged view of the orthophoto from December 16, 2021.

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The results for melt pond area presented in Fig. 6(e) and (f) indicate a marked difference. The pond area reveals a pronounced right-skewed distribution, with 8783 melt ponds (71% of the total) having areas smaller than 1 m^2, 11 316 melt ponds (91% of the total) having areas less than 10 m^2, and 12 147 melt ponds (98% of the total) having areas less than 50 m^2. The median melt pond area is notably smaller than the mean, as shown in Table III. This further indicates that small ponds are predominant among melt ponds. The roundness of ponds also reveals a pronounced right-skewed distribution [see Fig. 6(g)]. All melt ponds exhibit roundness values greater than 1. There are 1038 melt ponds with a roundness less than 2, and 11 360 melt ponds with a roundness greater than 2, constituting more than 92% of the total. The mean and median roundness of the melt ponds are both greater than 2 (see Table III). The results further demonstrate that melt pond shapes in the region exhibit non-circular characteristics, which are also intuitively illustrated in Fig. 6(h). Melt ponds with a roundness exceeding 4 account for 33%, demonstrating the presence of a significant number of irregular ponds in the region. Similar to length and width, the relationship between roundness, area, and probability density also exhibits a power-law nature.

B. Distribution of MPF on Landfast Ice Near Zhongshan Station

The constructed RFR model is used to retrieve MPF on landfast ice near Zhongshan Station and to monitor its evolution during the 2021–2022 summer season. After excluding the influence of clouds and cloud shadows, a total of nine Landsat 8 images are successfully employed in MPF mapping. Normalized Difference Water Index for ice ($NDWI_{ice}$) is used to obtain potential pixels containing ponds with a threshold of 0.12 [52].

Spatial distribution patterns of MPF during the melt season extending from early November to early March of the following year were examined (see Fig. 7). Melting features appeared at the edges of Nella Fjord, Clemence Fjord, Quilty Bay, Thala Fjord, and Wilcock Bay before November 8, 2021. MPF was high near the shoreline and decreased progressively as it extended outward. On November 17, 2021, surface meltwater receded rapidly and was exclusively confined to Wilcock Bay and Quilty Bay, where it retreated toward the coastline. A week later (November 24, 2021), there was a significant increase in MPF, with widespread melting observed around every bay and island. These strong melting features persisted until December 19, 2021, and MPF remained stable. On January 2, 2022, melting features retreated toward the coastline. Melt ponds within Thala Fjord and the McLeod Island region had almost entirely disappeared. Widespread melting occurred again two days later (January 4, 2022), accompanied by a sharp rise in MPF. After February 12, 2022, melt features far from the coast decreased significantly. However, melt features with lower MPF still existed near the coastline.

Focusing more closely on the Mirror Peninsula, the site of Zhongshan Station, regions with high MPF frequently appeared in Nella Fjord on the peninsula's western side

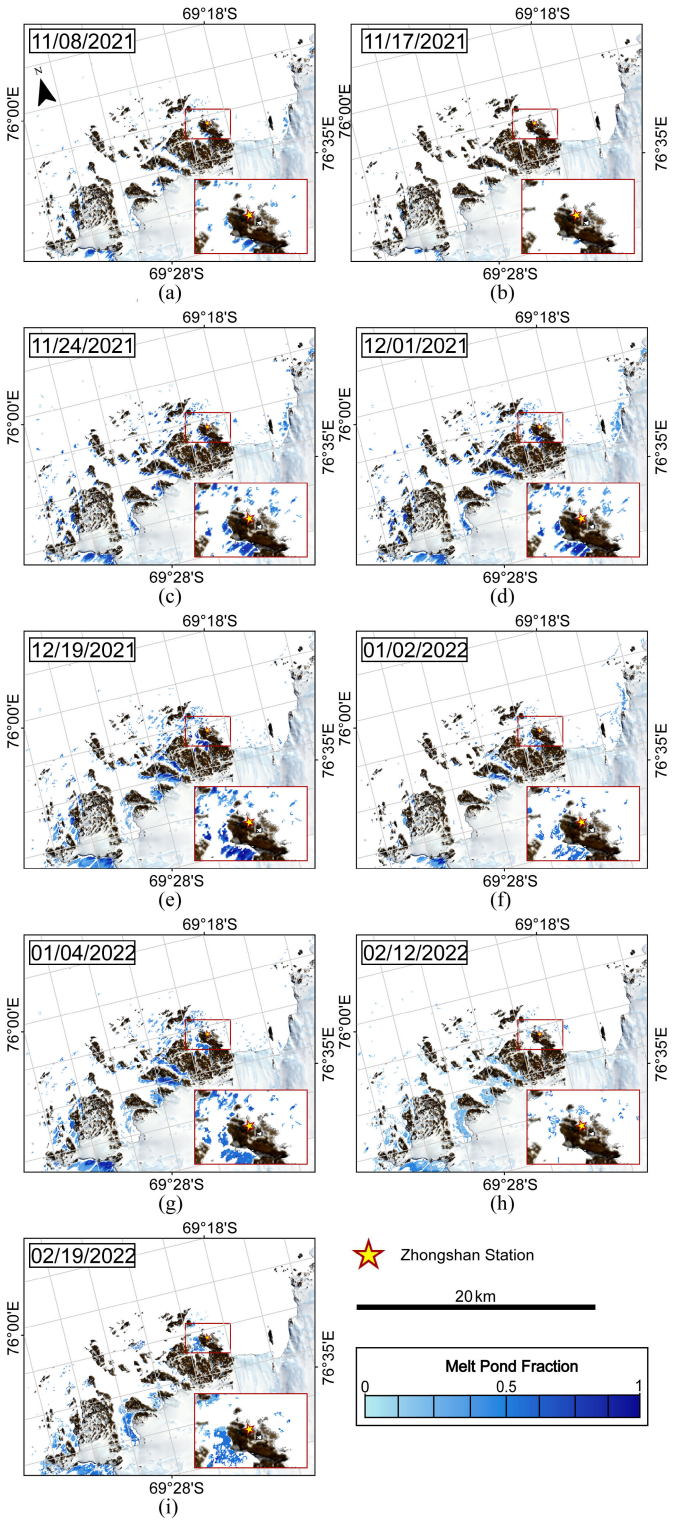


Fig. 7. MPF distribution on landfast ice near Zhongshan Station during the 2021–2022 melt season. (a)–(i) Correspond to MPF results derived from Landsat 8 imagery on different dates.

throughout the melt season. In contrast, melting features in Dalkoy Bay to the east were more dispersed and displayed lower MPF. During the early stage of melting (before mid-November 2021), meltwater around the Mirror Peninsula was limited, and MPF values were relatively low. From

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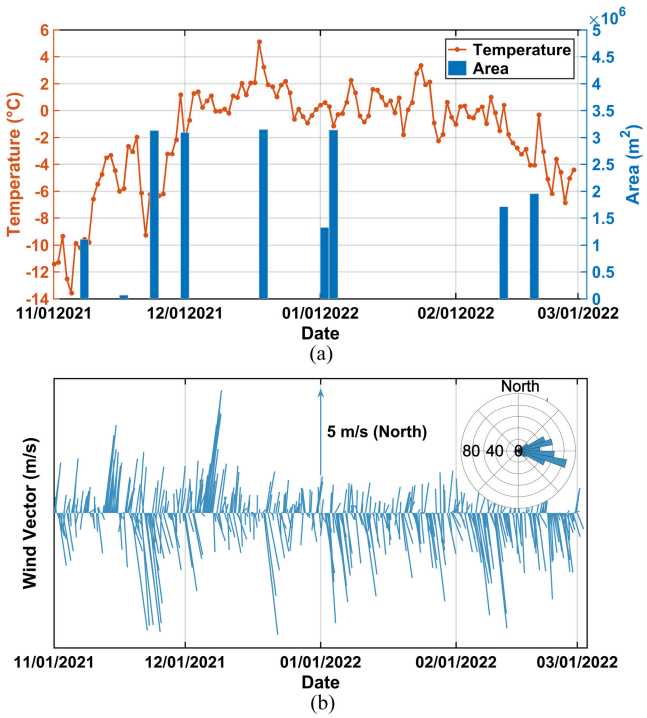


Fig. 8. Variations in melt pond area observed from satellite, as well as air temperature and wind field at Zhongshan Station from November 1, 2021, to March 1, 2022. (a) Temporal variation in melt pond area and surface air temperature. (b) Temporal variation in wind field.

mid-November to early January, melting features on the western side persisted and exhibited higher MPF. On the eastern side, the distribution of features fluctuated considerably, though their extent remained limited, and MPF values were lower. A comparison between MPF on February 12, 2022, and on February 19, 2022, showed that more melting features were present on the east side on the former date, whereas the latter showed a more pronounced concentration on the western side.

Fig. 8(a) shows the changes in pond area and air temperature in the region during the monitoring period. Overall, variations in melt pond area closely corresponded with surface air temperature measurements recorded at Zhongshan Station. The area increased rapidly early in the melt season, climbing from 1.11×10^6 m^2 on November 8, 2021, to about 3.15×10^6 m^2 by December 19, 2021, as the air temperature increased above the melting point. The reduction in melt pond area caused by intermittent refreezing (e.g., November 17, 2021, and January 2, 2022) was typically accompanied by a decrease in temperature. Following the peak of melt pond area in early January 2022, the landfast ice gradually refroze. Throughout the melt season, easterly winds predominantly influenced the region [see Fig. 8(b)]. This wind pattern is closely associated with the observed concentration of melt ponds along the western side of the coastline. Moreover, it was observed that during periods of high MPF (November 24 to December 19, 2021), wind speeds were relatively high, with a mean value of 6.09 m/s. In addition to the melt pond monitoring results for the 2021–2022 melt season, the RFR model was further applied to Landsat 8 data acquired during the 2017–2018 melt season (see Table SI). The retrieved MPF exhibited similar spatial patterns and seasonal characteristics (see Figs. S3 and S4).

TABLE IV
RISK STANDARDS DETERMINED FROM MPF AND NDWI$_{ICE}$

Level	MPF	NDWI$_{ice}$	Explanation
No risk	0	0-0.12	No coverage, No water
Low risk	0-0.1	0.12-1	Low coverage, All depth
	0.1-0.2	0.12-0.25	Medium coverage, Shallow water
Medium risk	0.1-0.2	0.25-1	Medium coverage, Deep water
	0.2-0.3	0.12-0.25	High coverage, Shallow water
High risk	0.2-0.3	0.25-1	High coverage, Deep water
Extreme risk	0.3-1	0.12-1	Extreme coverage, All depth

C. Melt-Induced Risk Assessment of Transportation on Melting Landfast Ice

To assess the melt-induced risk of transportation on landfast ice, we consider the MPF and NDWI$_{ice}$ on the ice surface. Based on MPF derived from Landsat 8, MPF is divided into no coverage (0), low coverage (0–0.1), medium coverage (0.1–0.2), high coverage (0.2–0.3), and extreme coverage (0.3–1). The melt pond depth can be reflected by NDWIice [52] and is categorized into three classes: no water (0–0.12), shallow water (0.12–0.25), and deep water (0.25–1). Combining MPF and NDWI$_{ice}$, we grade the melt-induced risk of ice surface transportation (see Table IV).

During the first monitoring (November 8, 2021), risk levels were dominated by extreme and low categories, with extreme risk concentrated along the western Ingrid Christensen Coast (see Fig. 9). In the vicinity of Zhongshan Station (western Mirror Peninsula), clustered risk zones were mainly distributed near the coastline, while the eastern and northern parts of the peninsula remained largely risk-free with only sporadic low-risk patches. On November 17, 2021, extreme and high-risk areas had largely disappeared, leaving only scattered low- and medium-risk patches along the western coast, whereas the eastern region remained risk-free. Only minimal low-risk areas persisted around Zhongshan Station. From November 24 to December 19, 2021, melt-related risk intensified significantly. Extensive extreme- and high-risk zones developed along the western coastline,

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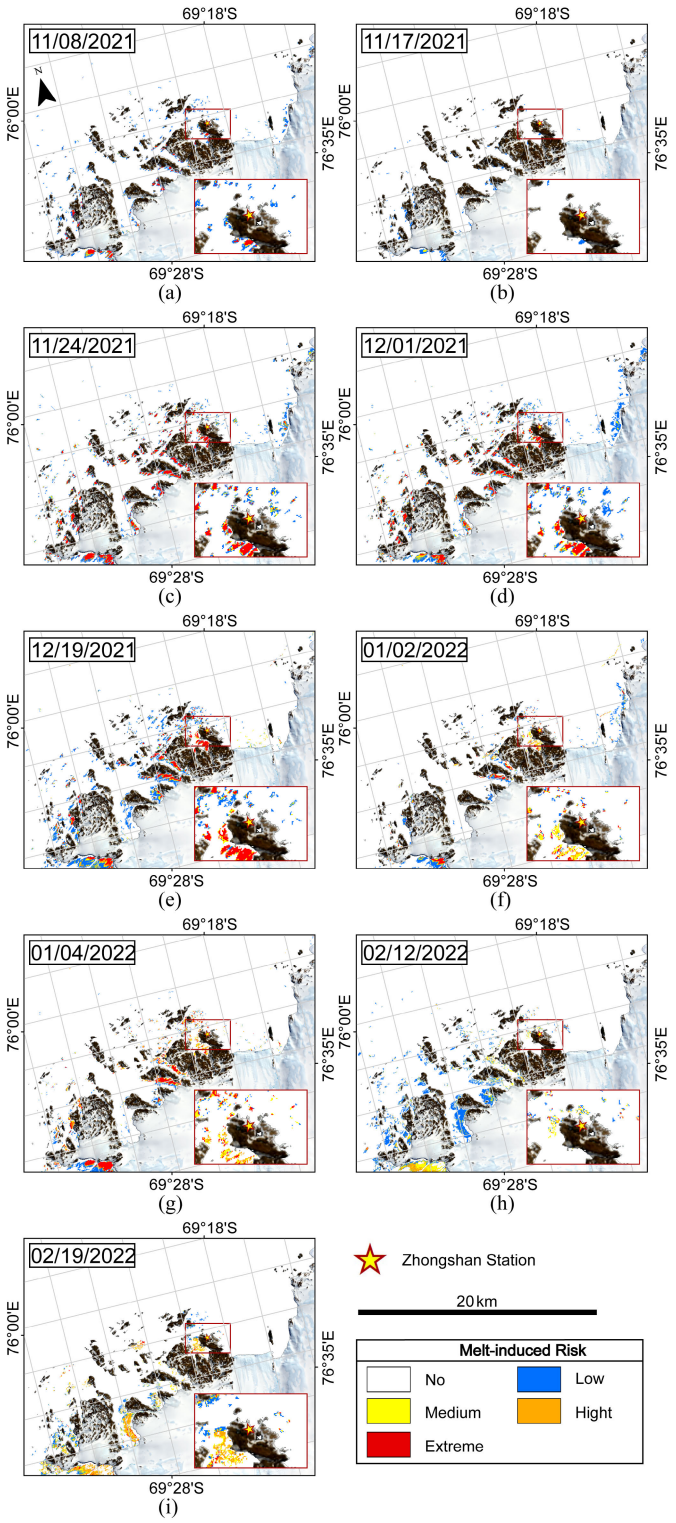


Fig. 9. Thematic maps of melt-induced risk over melting landfast ice near Zhongshan Station during the 2021–2022 melt season. (a)–(i) Correspond to the risk assessment results on different dates.

particularly between Manning Island and McLeod Island, while Nella Fjord (western Mirror Peninsula) became dominated by extreme risk. Medium and low-risk areas also expanded northeastward, gradually encircling the Mirror Peninsula. On January 2, 2022, overall risk levels decreased, with extreme risk persisting mainly near Fisher Island and the western coast of Wilcock Bay. Nella Fjord (west of Zhongshan Station) transitioned to medium and high risk, while scattered risk areas in eastern Dalkoy Bay diminished. On January 4, 2022, risk areas slightly re-expanded around Fisher Island, Quilty Bay, and Wilcock Bay, with moderate intensification near Zhongshan Station, where medium and high-risk conditions remained dominant. On February 12, 2022, western regions of Fisher Island showed renewed expansion of low- and medium-risk areas, while eastern regions contracted. Risk areas west of Zhongshan Station disappeared, whereas low- and medium-risk patches persisted to the east and north. Medium-risk areas expanded to dominate the western sector of Zhongshan Station on February 19, 2022, while low-risk zones were compressed into a few isolated pockets.

V. Discussion

The acquisition of UAV imagery, classification of melt ponds, and estimation of MPF all introduce uncertainties into the research. UAV imagery was collected from Nella Fjord, Clemence Fjord, and Quilty Bay, situated west of Zhongshan Station, through three flight missions (see Fig. 2). Independent validation using data from Flight 3 demonstrated good model performance. However, the representativeness of the data remains limited, and the applicability of the algorithm across the entire Antarctic region requires further evaluation. In addition, the extreme climatic conditions in Antarctica pose significant challenges to the deployment of real-time kinematic (RTK) systems required for high-precision UAV geolocation. This limitation may induce discrepancies in the alignment between Landsat 8 and UAV-captured imagery, thereby introducing uncertainties in subsequent analytical estimates. However, visual interpretation reveals remarkably consistent feature matching between the two observation systems (see Fig. S1), which is attributed to sufficient satellite positioning accuracy. Meanwhile, existing literature indicates that even without RTK, the localization errors remain substantially smaller than a Landsat pixel (30 m), exerting a negligible impact on the validity of this study's conclusions [53], [54].

Although the overall classification accuracy exceeds 90% in melt pond extraction from UAV imagery, a small number of pixels are still misclassified. These misclassifications may introduce minor deviations in the statistical analysis of melt ponds. In addition, UAV-derived results serve as ground truth for the development and assessment of the algorithm, indicating that any classification inaccuracies may propagate uncertainties throughout both the algorithm construction and validation stages. We integrated UAV-based classification errors and RFR estimation errors to derive spatial distributions of MPF estimation uncertainty for Landsat 8 imagery (see Fig. S5), showing that the resulting uncertainty remains limited, with a maximum pixel-level RMSE of only 0.16. For satellite imagery, since melt ponds are a form of surface water, this study employs a threshold-based method ($NDWI_{ice} > 0.12$) to extract pixels associated with melt ponds. In practical applications, this index proves inadequate for effectively differentiating melt ponds from water-filled surface cracks. Such conditions are likely to cause an overestimation of MPF within

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the study area [26]. Because cracks are typically narrower than the spatial resolution of Landsat 8 pixels, such overestimation is generally difficult to quantify [55], [56]. Through visual interpretation, we identified cracks large enough to contain visible water, and found that the maximum area of such misclassified features was only 5.75×10^6 m^2 (January 2, 2022), accounting for approximately 2% of the observed melt pond area (see Fig. S6). Therefore, we consider their impact on the overall results of this study to be negligible.

This study quantifies the length, width, area, and roundness of melt ponds based on high-resolution UAV data. The results indicate that melt ponds are mainly composed of small, noncircular features, with only a few highly irregular large ponds, which is consistent with observations previously reported in the Arctic [57], [58], [59], [60]. This pattern may arise from uneven winter snow and ice topography, such as snow dunes and pressure-ridge keel remnants, which produce numerous small, irregular depressions that serve as isolated meltwater catchments and bias pond development toward small, noncircular shapes [61], [62]. Moreover, as meltwater begins to percolate early through cracks and brine-channel networks, lateral coalescence is inhibited, so that only a few ponds grow to large sizes, and those that do exhibit highly irregular, noncircular outlines once a critical percolation threshold is surpassed [63]. At the same time, we find that the distribution of melt pond areas follows a power-law pattern, consistent with the findings of Perovich and others [63]. However, the power-law coefficients derived in this study are lower than those reported in earlier research [48], [64], [65]. This difference is attributable to the high resolution (5 and 11 cm) of the UAV imagery, which captures many small melt ponds ignored in prior studies and likely affects the power-law fitting parameters [65], [66].

Two MPF retrieval approaches were developed and evaluated, and the comparative results indicate that the conventional LSU-based method exhibits clear limitations when applied to subpixel-scale melt pond mapping over Antarctic landfast ice. Although the LSU algorithm demonstrates a certain degree of stability within the study region, the MPF estimates exhibit considerable overestimation and modest correlations ($r < 0.65$), and thus do not meet the required accuracy standards. These deficiencies are likely attributable to the complex surface and radiative transfer conditions in the Antarctic coastal environment. Multiple scattering within heterogeneous ice–water–snow mixtures, specular reflection from smooth melt pond surfaces, the presence of thin ice layers, and strong spatial heterogeneity in surface properties can all lead to nonlinear spectral responses that are not adequately captured by linear models [24], [32], [67], [68], [69]. In contrast, the RFR model, trained using joint UAV and satellite observations, demonstrates substantially improved retrieval accuracy and enhanced generalization capability. As an ensemble learning method, RFR is well suited to handling nonlinear relationships, feature interactions, and noise-contaminated inputs, making it more robust under complex surface conditions. The additional stratified evaluation over dust-affected and dust-free melt pond regions shows minor performance differences, further

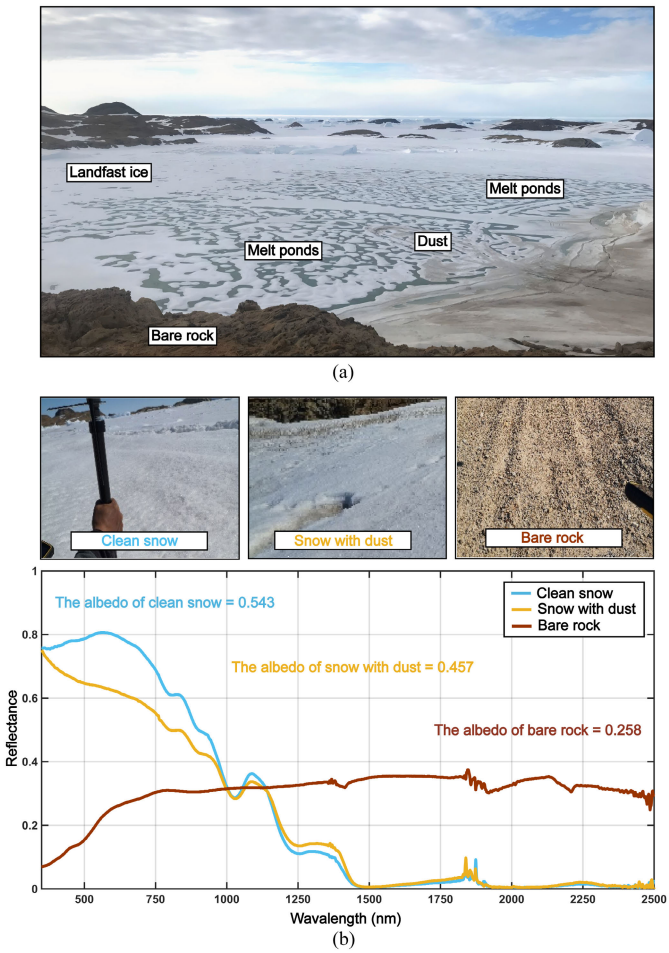


Fig. 10. Field-observed surface features and their spectral information near Zhongshan Station. (a) Melt ponds associated with dust originating from bare rock were observed west of Zhongshan Station. Image credit: Lei Zheng, December 17, 2021. (b) Spectral reflectance of clean snow, snow with dust, and bare rock was collected near Zhongshan Station. Image credit: Xi Zhao, December 28, 2019.

confirming the robustness of the proposed approach under varying surface conditions (see Fig. S7).

Satellite monitoring based on the RFR algorithm indicates that MPF near Zhongshan Station is concentrated along the western part of the bare rock coastline, increasing with proximity to the coast, while other regions show sparse distribution and lower MPF. This pattern is closely associated with bare rock exposure and prevailing wind conditions. Bare rock (albedo = 0.258, see Fig. 10) enhances solar absorption, accelerating adjacent ice melting [70]. Antarctic field observations confirm that supraglacial ponds tend to form near bare rock outcrops due to melt-albedo feedback [71]. We infer that the prevailing easterly winds in the study region [see Fig. 8(b)] transport dust from exposed rock surfaces to leeward ice areas, thereby reducing surface albedo along the western coastline and further accelerating ice melt [72], [73], [74], [75]. This interpretation is also confirmed by the field images and measured surface spectral information collected near Zhongshan Station (see Fig. 10). The albedo of snow with dust is approximately 0.1 lower than that of clean snow, which

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substantially enhances radiative absorption. Furthermore, melt pond area is temporally correlated with air temperature [see Fig. 8(a)], peaking and declining synchronously during the melt season, underscoring the temperature's role in driving the evolution of melting [27], [76].

Melt-induced risk assessment reveals that high and extreme risk zones cluster spatially west of the Ingrid Christensen coastline. Notably, intense melting in this region results in the formation of extensive risk clusters. Nella Fjord, located west of Zhongshan Station, frequently undergoes large-scale melting during summer, with risk levels generally remaining above medium. In contrast, eastern/northern ice areas exhibit dispersed, lower-risk patches with limited coverage. In the early melt stage before mid-November, the overall risk is the lowest, and there is even a period of refreezing. Based on these findings, we recommend that summer transport and unloading operations on landfast ice should be planned away from the western area of Zhongshan Station. Operations should ideally conclude before mid-November to leverage minimal melting conditions and mitigate potential risks.

At present, MPF retrieval based on optical satellite imagery, such as Landsat 8, is constrained by cloud cover, illumination conditions, and sensor revisit cycles, which limit monitoring frequency during the melt season. To investigate whether microwave data can enhance the temporal continuity of melt pond observations, we followed the approaches used in previous microwave-based melt pond studies [17], [19], [20], [21], [22] and conducted an exploratory analysis over the study region using Sentinel-1 C-band SAR data (see Text S2). The results indicate that there is no evident correlation between the C-band backscatter coefficients, the polarization ratio, and MPF across the study area (see Fig. S8). The weak correspondence is likely related to strong surface heterogeneity and complex scattering mechanisms, including the influence of wind-induced surface roughness, variations in the physical state of melt ponds (e.g., partial freezing or thin ice cover), as well as differences in ice type (e.g., multi-year versus first-year ice) and sensitivity to SAR incidence angle [16], [17], [19], [20], [21], [22]. While SAR observations offer important advantages in terms of all-weather and illumination conditions, their direct application for quantitative MPF retrieval in this region remains challenging. Future research efforts should integrate more high-precision UAV observations from different periods and regions, and combine multiple optical satellite datasets (e.g., Sentinel-2) to develop more robust monitoring models, thereby improving retrieval accuracy and better capturing the rapid spatiotemporal dynamics of melt ponds. Meanwhile, the existing risk assessment framework primarily addresses surface melting, neglecting important factors like sea ice thickness and ice cracks. Future studies should integrate multi-source datasets to develop a more comprehensive assessment framework and provide a solid data foundation for polar scientific research route planning.

VI. Conclusion

This study integrates UAV and satellite remote sensing to investigate, for the first time, the characteristics of melt ponds on Antarctic landfast ice near Zhongshan Station. High-resolution UAV-derived data reveal that the melt ponds in this region are small (median area = 0.34 m^2) and irregular in shape (mean roundness = 4.97). To achieve broader-scale melt pond monitoring in Antarctica, we propose an RFR model for retrieving MPF from Landsat 8 imagery. Validation against UAV data shows that the model outperforms traditional LSU methods, achieving an r of 0.82 and an RMSE of 0.10. Dynamic monitoring reveals that during the 2021–2022 melt season, melt ponds were highly concentrated on the west of the bare rock coastline, with higher MPF values observed closer to the shore, likely due to persistent easterly summer winds transporting dust westward. At the same time, we find that the temporal evolution of melt ponds is significantly affected by temperature. Furthermore, we establish a melt-induced risk assessment framework that integrates MPF and pond depth, aiming to directly support Antarctic expedition route planning and ensure the safety of ice surface operations. High melt-induced risk areas are primarily concentrated in the ice region of Nella Fjord, located west of Zhongshan Station. We recommend avoiding this region and completing ice surface transport operations before mid-November to minimize potential risks.

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